

# TTTV (Taste the TV): Taste Presentation Display for “Licking the Screen” using a Rolling Transparent Sheet and a Mixture of Liquid Sprays

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## ABSTRACT

Reproducing the taste of food and beverages as a media technology is an emerging problem of commercial value and entertainment utility. An already developed taste reproduction system uses electrolyte containing gels controlled by electric current to reproduce taste by ion-phoresis. In this study, an alternative taste reproduction mechanism was proposed that sprays a mixture of liquids on a rolling transparent sheet placed on a screen that displays the image of the food item under consideration for reproduction of its taste.

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices.

## KEYWORDS

taste display, taste reproduction, taste sensor, liquid spray

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## 1 INTRODUCTION

I invented the Norimaki Synthesizer [1], an ion-phoretic taste display that can replicate any arbitrary taste, and they developed the idea of linking it to taste sensors [2]. The Norimaki Synthesizer consists of five gels containing electrolytes that can produce the five basic tastes and the attached electrodes. While holding the electrodes to apply the gels against the tongue, the amount of electrolytes in contact with the tongue can be varied, allowing adjustment of the balance between the five basic tastes to produce any given taste. Furthermore, the taste of any food or drink can be reproduced and tasted by calibrating taste reproduction based on values acquired through taste sensors. This study also used

taste sensors [3] [4] [5] for the development of a taste display system. However, rather than using ion-phoresis, it proposes a taste reproduction mechanism that uses a mixture of liquid sprays.

## 2 SYSTEM

### 2.1 Taste Display

The proposed system allows its user to taste the flavor of the food displayed on the screen by directly licking it. As shown in Fig. 1, the proposed taste display has a size of 420 mm in width, 500 mm in height, and 800 mm in depth, and weighs about 20 kg. The upper part of the system is equipped with 10 spraying devices, which spray and mix the liquid solutions onto a transparent sheet which rolls out toward the user. Beneath the sheet is a 13.3-inch LCD screen displaying the image. When licking the screen, the user is able to taste the mixture of liquids sprayed on the sheet, with different flavors depending on the contents shown on the screen. This allows the user to experience the taste of the food displayed on the screen by directly licking it.

The image on the left side of Fig. 2 is a single unit of the spraying mechanism. It is implemented by modifying a double action type airbrush. It had a tank capacity of 40 mL for each type. The amount of liquid was adjusted by fixing the air pressure and controlling the spraying time (the minimum discharge amount was 0.01 ml). By gradually spraying 10 types of pumps, it is possible to mix their flavors without the need for stirring.

The image on the right side of Fig. 2 shows the detachable rolling apparatus. The sheet roll is 30 m long oriented polypropylene (OPP), with a thickness of 0.04 mm; the roll is discarded after use. The spray apparatus and rolling apparatus were both controlled by serial communication (UART 115200bps) from a Windows PC connected to a 13.3-inch LCD monitor (without touch function) via HDMI that was placed beneath the sheet. A sheet sprayed with liquid is shown in Fig. 3. The liquids used were clear and colorless, so that the image was clearly projected with its original colors. However, there was a reduction in the resolution.

### 2.2 Taste sensors and calibration

There are eight types of taste presentation solutions that fill each tank, including no taste, as shown in Table 1. There are two extra tanks at the moment for potential future work.

These liquid solutions are same as the basic taste sample solutions used in the calibration of each sensor of the taste sensor TS-5000Z [3] [4] [5]. Each taste has a concentration approximately 10 times the human perceptual threshold, and the taste of most foods and drinks that a person eats does not exceed this level of intensity (except when licking seasonings directly). Therefore, in

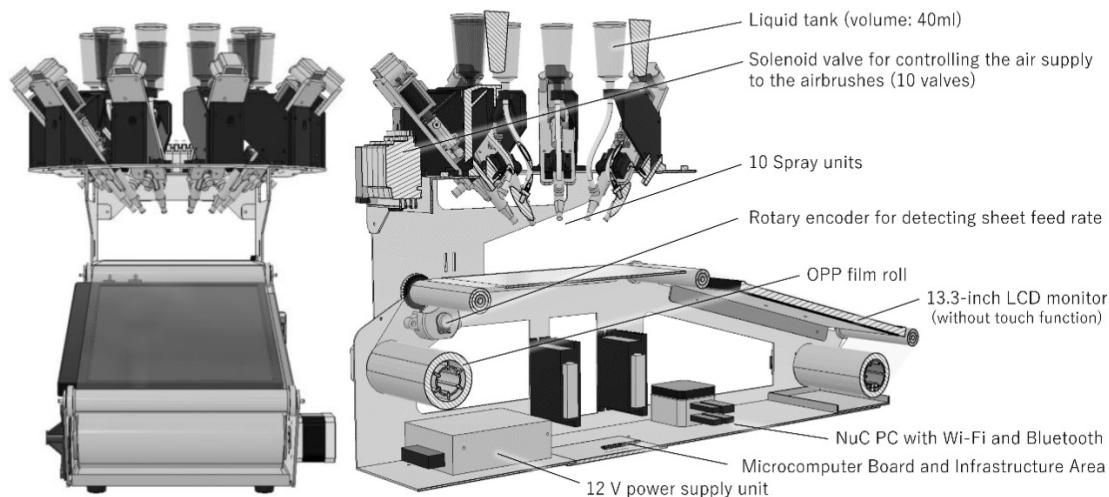
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**Figure 1: (Left) External view, (Right) Internal mechanism**



**Figure 2: (Left) Spray unit, (Right) Rolling apparatus (TEST has been written to indicate the presence of transparent sheet)**

**Table 1: Tastes presented and solutions used**

| Presented taste    | Solution                                   |
|--------------------|--------------------------------------------|
| Salty              | 300mM Potassium chloride aqueous solution  |
| Sour               | 0.3mM Tartaric acid aqueous solution       |
| Umami              | 10mM Monosodium glutamate aqueous solution |
| Bitter (medicinal) | 0.1mM Quinine aqueous solution             |
| Bitter (culinary)  | 0.01vol% Iso-alpha acid aqueous solution   |
| Astringent         | 0.05wt% Tannic acid aqueous solution       |
| Sweet              | Sucrose aqueous solution                   |
| No taste           | Purified water                             |

principle, the combination of these solutions can be considered to reproduce the taste of most foods and drinks. There are five basic tastes, but the taste sensor TS-5000Z can also measure “astringency,” which is not included, as well as two other types of taste: bitterness as found in medicinal products and bitterness as found in foods, for a total of seven tastes as the output data. The proposed system reproduces these data by combining these seven solutions. Calibration with the taste sensor was similar to that in [2]. The taste sensor outputs a relative value when compared to the liquid. To obtain the

data for the proposed system, the taste of food and drink may be quantified using an aqueous solution in which all seven solutes are dissolved as a comparison target. By back calculating the intensity values of each taste type, a concentration converted to each of the above reference solutions can be obtained. The proposed system can reproduce tastes by mixing the sprays of liquid solutions based on these values. Although Satoh et al. [6] proposed a mechanism based on genetic algorithm to determine the mixture proportions of



**Figure 3: Screen with sheet that has been sprayed with the liquids**

the component solutions, the system proposed here only produces the concentrations for reproducing the value of each parameter.

### 3 DISCUSSION

The system proposed in this paper has many advantages over the system in [2] (hereinafter referred to as ion-phoretic taste display) in which the Norimaki Synthesizer [1] is connected with the taste sensors. First, it can express a different range of tastes. The ion-phoretic taste display expresses taste intensity via electrical adjustment of the electrolytes dissolved in the gels. In absence of electric current, the user feels the gel’s taste at full intensity. Applying electric current brings the taste closer to tastelessness. However, when further intensifying the gel taste, it will not be possible to make the taste completely tasteless, no matter how much of an electrical current is applied. Therefore, the ion-phoretic taste display has limitations on the taste intensity of gels. In fact, based on [1] and [2], the salty gel, for example, uses 1% NaCl, which is not enough to reproduce the salty taste of most foods and drinks. In contrast, the spray mixing method proposed in this paper produces “no taste” by not spraying solutions, or by spraying tasteless purified water, while having almost no upper limit on taste intensity. Certainly, the concentrations shown in Table 1 are the upper limit of taste intensity, but an increased range of intensity expression is possible by simply increasing the concentrations of the solutions in the tanks, even up to the point of saturation. Furthermore, the spray mixing method does not cause any electrical stimulation, avoiding any such discomfort.

In addition, the proposed system succeeded in expanding the range of possible tastes, which was limited to the combination of the five basic tastes in the ion-phoretic method, to seven tastes that can be measured by taste sensors. In the ion-phoretic method, the electrolytes must be used; therefore, to express, for example, sweetness, it is necessary to find a way to use glycine. However, the spray mixing method does not require such devices; therefore, sweetness can be simply expressed using sucrose.

Furthermore, taste experience itself is quite different in the two methods. With the ion-phoretic method, the taste can only be experienced at the contact surface when the gel is placed on the tongue. However, with the spray mixing method, the taste is actually present right up to the moment of licking and swallowing.

Another advantage of the spray mixing method is its ability to accurately calibrate the taste. In the ion-phoretic method, taste was calibrated using data from evaluation experiments to determine the level of intensity people perceive. Moreover, since there are individual differences, therefore the parameters must be changed depending on who uses the device. However, in the spray mixing method, the taste of the mixed solution can be measured using taste sensors. In other words, it is possible to use taste sensors again to test foods or drinks whose taste is targeted for reproduction, as well as the mixed spray solution used to reproduce that taste, allowing for a quantitative comparison of whether two persons have the same taste with an objective numerical value. This makes it possible to consider a measurement method that further improves reproducibility.

However, in some respects, the spray mixing method is inferior to the ion-phoretic method. First, the spray mixing method does not allow for immediate adjustment of the taste. A delay of several seconds is unavoidable due to the time taken for spray mixing and sheet winding. In addition, it is somewhat difficult to perform dynamic taste changes, such as a “taste crossfade,” through spray mixing. However, it is possible to overcome this by improving the control accuracy to realize a beautiful “taste gradation.”

In certain aspects, spray mixing and ion-phoretic methods are equally capable. Taste effectors and taste equalizers that change the balance of the basic five or seven tastes work with either method. It is also feasible to adjust the intensity as desired, regardless of the method.

An area for improvement is the clouding and reduced resolution of the image due to droplets from the liquids, although even the ion-phoretic method would require the use of gel resulting in the same problem. This is an aspect of both methods that require improvement in future.

### 4 FUTURE WORK

The current system works on the principle of spray mixing in one place. However, in the future, it would be desirable to have a mechanism with a head that moves around, similar to an inkjet printer. Doing so would allow taste to be represented in a two-dimensional arrangement with distribution across the image.

The spray mixing taste display also has room for adding alcoholic and spicy tastes that cannot be measured by taste sensors. Because there is a commercially available ethanol concentration meter [7], it is possible to use it as a basis for spray-mixing ethanol. In addition, although spiciness is academically classified as a sensation of pain, it would be desirable to add it as a possible taste to reproduce the eating experience. In fact, there is a commercially available scoville meter [8] that makes it possible to measure the spiciness of food, and it would be ideal to use it to improve the system so that capsaicin polypyrrole glycol solution can also be spray mixed according to the measured values.

Although it is out of the main purpose, the proposed system can be used as an apparatus for seasoning. For example, if crackers are placed under the spray unit, they can be sprayed with various flavors such as cheese flavor, tomato flavor, liver pate flavor, honey flavor, and so on. I would like to explore the possibility of using the system as a device for gastronomy.

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